Carbon Debt Computing

A deadline-aware controller that books emergency carbon as debt and repays it in green windows

Challenge 9 · Individual project
[STUDENT NAME] · [REGISTRATION_NO]
Cloud computing

The problem

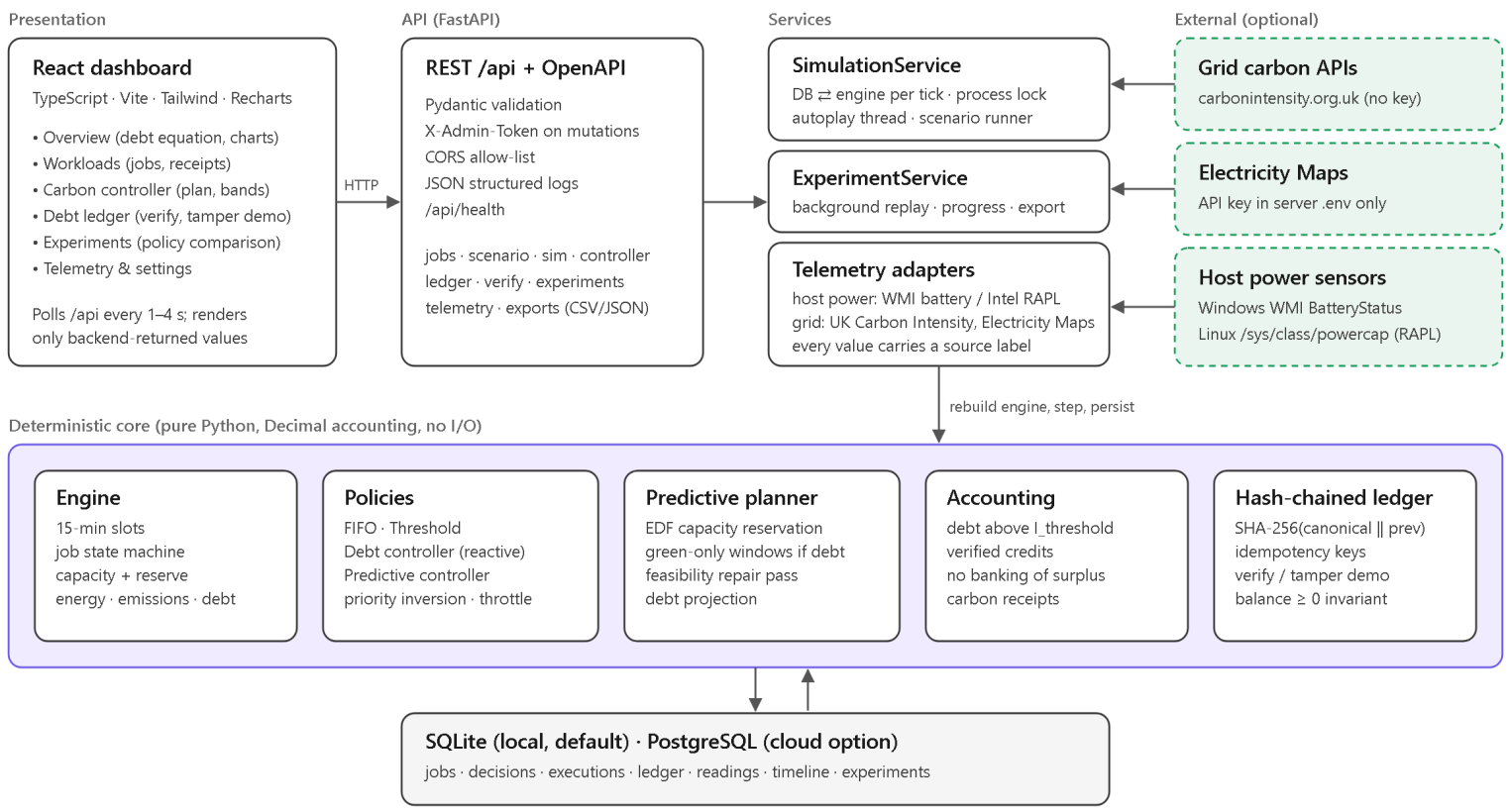
Emergency jobs, such as a hospital model failover or a flood-alert recompute, cannot wait for clean power. Carbon-aware schedulers defer flexible work, but they forget the unavoidable emissions and never make them up.

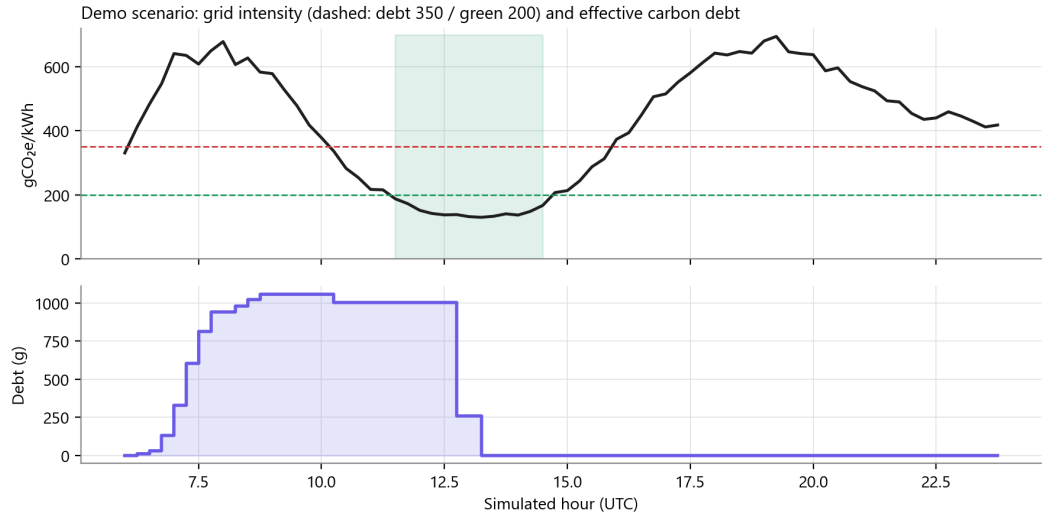
1,059 g (debt incurred) − **1,059 g** (verified credits) = **0 g** (outstanding)

The demo scenario, computed by the backend: two morning emergencies at 550–650 gCO_2e/kWh create debt, which is repaid by 13:30 with no missed deadline.

How it works

1. **Book the debt.** energy × max(0, intensity − 350) per 15-min slot, appended to a SHA-256 hash-chained ledger.
2. **Invert priority.** While debt > 0, flexible work runs only in green slots (≤ 200), ranked by repayment value instead of nominal priority.
3. **Throttle progressively.** Debt bands set background/normal capacity to 100/50/20%/paused. Never emergencies, always deadline-safe.
4. **Repay with verified credits.** Baseline recorded at deferral; credit = $E \cdot I_{baseline}$ − actual emissions; no duplicates, no banking.
5. **Predict.** An EDF capacity-reservation planner chooses feasible windows over the forecast and projects the debt path, raising throttling early.
6. **Prove it.** Ledger verification and a tamper demo, a carbon receipt per job, and replay of identical traces across policies.





Demo scenario (predictive controller). Debt rises during the morning peak and falls to zero as deferred jobs run in the solar green window.

Results — 10 seeds × 36 jobs

Policy	Emissions	Debt incurred	Final debt	Misses/run
FIFO	20.6 kg	5.38 kg	5.38 kg	0.2
Threshold	17.6 kg	2.96 kg	0.47 kg	1.3
Debt controller	16.9 kg	3.16 kg	0.07 kg	3.8
Predictive	**17.2 kg**	**2.88 kg**	**0.25 kg**	**1.3**

Against FIFO, the predictive controller cut emissions by **16.7%** and debt by **46.5%**, with deadline misses equal to threshold deferral. Forecasting cut misses from 3.8 to 1.3 per run compared with the reactive controller. The price is delay for flexible work (~6.8 h on average).

Telemetry

Live mode: Indian grid intensity from India Energy Atlas, the Windows process list (psutil), and measured CPU-package energy from the Windows Energy Meter (Intel RAPL), shared to each job by CPU time.

Engineering

FastAPI · SQLAlchemy · SQLite/PostgreSQL · React · TypeScript · Recharts. Decimal accounting, 76 automated tests, deterministic replay, Docker + Caddy HTTPS deployment.

Honest limits

Energy is estimated, the cluster is simulated locally, and the default grid curve is illustrative. Credits count avoided *future* emissions and never erase past ones. The hash chain is trustworthy only against an external head checkpoint.